

David Silveira

Software Engineer

Austin, Texas | djs3k@proton.me | linkedin.com/in/david-silveira-03921821b | github.com/daviewave

SUMMARY

Software Engineer with 5 years of experience taking AI-powered products from concept to production through full-stack development, system architecture, and technical leadership. Most recently led Wolverine, a self-healing cybersecurity knowledge-graph platform, from proof of concept to an agentic AI product on AWS Marketplace.

SKILLS

Programming Languages: Python, TypeScript, JavaScript, SQL, Bash, C, C++, R

Backend: Django, Django REST Framework, FastAPI, Flask, Celery, WebSockets, REST APIs, microservices

Databases & Query Languages: PostgreSQL, MongoDB, Redis, Neo4j, Amazon Neptune, openCypher, SPARQL

AI & Machine Learning: AWS Bedrock, LangChain, GraphRAG, agentic AI, LLM guardrails, NLP, Whisper

DevOps & Testing: GitHub Actions CI/CD, trunk-based development, Docker, Jenkins, QEMU, pytest, Playwright

Scripting & Automation: Bash and shell scripting, Linux, Makefiles, Python automation, scheduled GitHub Actions

Observability & Telemetry: OpenTelemetry, Prometheus, Grafana, Amazon CloudWatch, Core Web Vitals

Frontend: React, Next.js, Vite, TanStack Query, Redux, Tailwind CSS, Material UI, design systems

Cloud: AWS (Bedrock, ECS/Fargate, EC2, ECR, S3, CloudFormation, IAM, Route53), Terraform, Nginx

EXPERIENCE

Lead Software Engineer

Cyberhill Partners | Austin, Texas | April 2026 - Present

- Led Wolverine from proof of concept to AWS Marketplace launch, setting up the workflow its 20 contributors and AI coding agents share: a 117-target Makefile as the single entry point, trunk-based development with reviewed PRs, and coding conventions that keep the code self-documenting.
- Automated documentation with weekly Bedrock-powered GitHub Actions that move verbose code comments into docs/, log code findings to per-area backlogs, and sync both to Confluence in plain language, saving an estimated 10+ engineering hours a week of writing docs for non-technical readers.
- Made a core graph analytics workload on Amazon Neptune 3.6x faster (6.34 s to 1.78 s) by batching sequential round-trips into a few set-based queries, and fixed a scoring bug that counted partial coverage as complete.
- Designed a dual-database architecture separating the knowledge analytics graph database layer from the relational data storage layer, transitioning the backend from FastAPI to Django to leverage its structured ORM for PostgreSQL while supporting high-throughput asynchronous graph database queries through Django Ninja.
- Engineered real-time AI chat that streams LLM tokens over WebSockets, with Redis as the message layer that keeps them in order, plus GraphRAG that turns plain-English questions into graph queries (openCypher, then SPARQL after an LPG-to-RDF move) to reveal how security tools and frameworks relate.
- Partnered with design and leadership on a process that automates 75-85% of frontend work: a layered design system that splits shared base components (L1-L3, an npm package) from per-app compositions (L4-L5), and AI agents that compare each design to the live app and plan its implementation.
- Created test suites at every level of the Wolverine app and research console, from pytest and Vitest unit tests through smoke and regression tests to a no-mocks Playwright suite of 111 end-to-end tests split into critical and non-critical tiers, plus contract tests that run after every graph update.
- Designed GitHub Actions CI/CD for trunk-based development, with test suites gating each stage and work split into parallel jobs and balanced test shards to cut run time, ending in automated AWS Marketplace releases with gated promotion, automatic rollback, and image digest checks.
- Codified lab environments in Terraform, hardened the 48-resource CloudFormation template customers launch, and built observability with OpenTelemetry, Prometheus, Grafana, and CloudWatch alarms.
- Architected a Django and React research console, replacing a Streamlit prototype, that researches security tools and adds them to the graph: Tavily search plus 4 LLM families on Bedrock, a state machine that holds every result for CISO approval (human in the loop), and support for LPG or RDF graphs.

Full Stack Software Engineer II

Ultra Intelligence & Communications | Austin, Texas | January 2026 - April 2026

- Split a monolithic Django and React application into independent microservices, writing a container file for each service and updating the Jenkins CI/CD scripts to build and deploy them separately.

- Helped migrate a legacy UI to React and TypeScript with Material UI and Styled Components, introducing feature-scoped Redux stores so each feature owns its state and the codebase stays maintainable.
- Separated real-time from non-real-time data in the company's flagship product, moving features that did not need live data into Django apps on PostgreSQL and reworking the C structs and C++ code on the real-time path.

Founder & Lead Engineer

Easy Covers Software | Wilmington, Massachusetts | May 2023 - January 2026

- Founded an AI-powered recruiting platform and shaped its features through extensive user research with recruiters, identifying their biggest pain points and building the product around them.
- Developed the product from scratch as containerized Django, Flask, and Next.js microservices on PostgreSQL behind a reverse proxy, deploying them to the cloud in separate dev and demo environments.
- Implemented transcription for recruiter phone calls placed through Twilio, using Kaldi recipes and OpenAI's Whisper model for speech recognition, and generated structured call notes automatically with LLMs.
- Designed machine-learning match scores that rated each candidate's fit for a role, feeding in the notes from recruiter calls as extra context so scores reflected what candidates said, not just their resumes.
- Wrote a 25-page business plan and pitched the product to potential customers and investors, earning meetings with venture capitalists; converted the LLC to a corporation.

Software Engineer (Contract)

Skaion Corporation | North Chelmsford, Massachusetts | February 2024 - January 2025

- Worked with company owners and project directors to help guide technical direction on a multi-million-dollar government R&D project, and briefed 3 Department of Defense stakeholders, helping secure more funding.
- Developed a full-stack Django and React application for a fully offline, no-cloud environment, with interactive 2D and 3D graph visualizations, packaged for deployment with Docker and Bash scripts.
- Refactored the Django backend into a lightweight command-line tool of modular Python scripts as the project's requirements changed, so each processing step can run and be reused on its own.
- Generated QEMU virtual machine commands with an offline AI service running a local GPT4All model instead of a cloud API, queuing requests through Celery and Redis to stay within strict memory limits.

Software Engineer

Forum Systems | Needham, Massachusetts | February 2022 - May 2023

- Joined Quantum Sim, an AI healthcare app, as its first external engineer before it spun out of Forum Systems as its own company, building its Django and React stack and partnering with 2 executives on ML-driven risk analysis.
- Developed a natural language processing (NLP) pipeline that extracts text from PDF contracts and analyzes it with Word2Vec embeddings, Transformer models, and GPT to compute a risk score for each contract.
- On a 4-person team, built an NLP healthcare benefits encoder and an ML product analysis engine that helped win Blue Cross Blue Shield follow-on funding to hire 3 developers, then onboarded them with 25+ shell shortcuts.

Undergraduate Research Assistant

Kwak Laboratory | Amherst, Massachusetts | January 2020 - June 2021

- Created an R pipeline that turned participants' free-text responses to 2018 midterm election prompts into structured data and classified each by emotion (anger, fear, neutral), showing how framing words shaped voter sentiment.

EDUCATION

Bachelor of Science in Psychology

University of Massachusetts, Amherst | 2017 - 2021

Neuroscience Concentration, Minor in Computer Science

Certificate in Web Development

Columbia University | 2021 - 2022

Software Engineering Boot Camp